

HEALTH AND MEDICINE

Rapid and robust diagnosis of bloodstream infections by single-cell analysis

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Sepsis, a common consequence of bloodstream infection (BSI), is a leading cause of global mortality and morbidity. Diagnosis is challenging and slow due to extremely low pathogen loads and reliance on time-consuming blood culture, necessitating empiric therapy that worsens outcomes and antimicrobial resistance. Here, we introduce sedimentation-assisted tandem rocking and enrichment for analysis and monitoring (STREAM), a platform that efficiently isolates and enriches pathogens near their natural doubling times directly from whole blood. By integrating STREAM with single-cell analysis, we achieved rapid and robust diagnosis of BSI from whole-blood samples. Using molecularly barcoded sequential fluorescence in situ hybridization, pathogen identification in 104 positive blood culture samples showed 96.15% concordance with clinical laboratory results. Gel-based single-cell antimicrobial susceptibility testing yields 97.96% essential agreement and 93.9% categorical agreement across 219 drug-dose combinations. The platform enabled comprehensive BSI diagnosis directly from whole blood, achieving a detection limit as low as 0.1 to 1 CFU/ml and delivering full diagnostic results within 6.75 to 17 hours.

INTRODUCTION

Bloodstream infections (BSI) are a leading cause of morbidity and mortality, imposing substantial healthcare and economic burdens (1). Clinically defined by the presence of pathogens in the blood, BSI often progresses to life-threatening sepsis, a dysregulated immune response causing systemic inflammation, organ failure, and death (2, 3). Rapid recognition of causative pathogens and early initiation of effective antimicrobial therapy are critical for patient survival. Delays in appropriate antibiotic therapy worsen clinical outcomes and increase mortality, while inappropriate initial treatment independently elevates the risk of death (4, 5). However, the gold-standard blood culture method requires 2 to 7 days to deliver definitive results. In the absence of timely identification of causative pathogens and their antimicrobial susceptibility, current guidelines recommend prompt empiric treatment with broad-spectrum antibiotics (6), a practice that can lead to undertreatment, worsen patient outcomes, and accelerate antimicrobial resistance.

BSI diagnosis is challenging due to extremely low pathogen loads, complex blood composition, broad pathogen diversity, and nonspecific clinical presentations (7, 8). The current gold-standard blood culture method requires enrichment of bacteria to detectable levels over 1 to 5 days, depending on microbial growth rates. Positive cultures are then subcultured to obtain isolates for pathogen identification (ID) and antimicrobial susceptibility testing (AST), which adds another 1 to 2 days. Rapid molecular diagnostic tests and single-cell analysis techniques have shortened detection times from days to hours (8–14); however, most still rely on time-consuming positive blood cultures. Culture-free methods using direct metagenomic next-generation sequencing, such as the Karius test, detect cell-free

microbial DNA directly from serum or plasma and can improve sensitivity for samples containing fastidious, slow-growing, or antibiotic-exposed organisms. However, these approaches are costly, have relatively long turnaround times, and do not provide phenotypic antimicrobial susceptibility results. Recently, direct analysis of whole blood through pathogen isolation has gained traction as a means to bypass lengthy culture steps (15–19). For instance, direct pathogen isolation from whole blood has been demonstrated using nanomagnetic beads coated with apolipoprotein H, enabling comprehensive diagnosis within approximately 13 hours and a limit of detection of 4 colony-forming units (CFU)/ml (15). However, isolation techniques that fully bypass culture are constrained by limited capture efficiency, increasing the risk of false negatives, especially in low-pathogen-load samples (<5 CFU/ml). Clinically important BSIs often contain very few bacteria per milliliter of blood, and episodes with ultralow loads (0.1 to 1 CFU/ml) can account for a substantial proportion of cases (20–23). Therefore, there remains an urgent need for rapid and robust diagnostic technologies capable of delivering actionable results, thereby enabling precision therapy and improving survival in BSI.

In this study, we present a BSI diagnostic platform that rapidly and robustly performs pathogen ID and AST directly from whole blood. We introduce sedimentation-assisted tandem rocking and enrichment for analysis and monitoring (STREAM), a technique that enables short periods of exponential bacterial growth while simultaneously isolating pathogens from ultralow-load blood samples. The system is sampled at regular intervals to detect early bacterial growth and perform pathogen ID, while continuous culture ensures sufficient growth to avoid false negatives and maintain diagnostic robustness. By integrating single-cell pathogen ID using molecularly barcoded sequential fluorescence in situ hybridization (MB-seqFISH) and phenotypic single-cell AST (24–26), the platform delivers comprehensive results in as little as 6.75 hours and reliably detects bacterial loads on the order of 0.1 CFU/ml, approaching the limit imposed by the sampled blood volume (~10 ml). We validated platform performance using more than 100 clinical blood culture samples and whole blood spiked with clinical isolates, demonstrating a rapid, sensitive, and

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high-throughput solution for BSI diagnosis. Our platform offers a powerful tool to guide evidence-based antibiotic therapy, enabling timely and targeted treatment within the clinically critical window and improving the management of BSI.

RESULTS

Design and overflow of our platform

Our single-cell diagnosis platform integrates three complementary modules to enable rapid, sensitive, and high-throughput BSI diagnostics directly from whole blood. The system is designed for an average sample-to-result time of under 9 hours, compared with the multiday workflow required for conventional blood culture (Fig. 1A). First, STREAM simultaneously enriches bacteria while clearing blood cells (Fig. 1B). When whole blood is mixed with a customized dextran-culture broth formulation and gently rocked, red blood cells (RBCs) aggregate into large rouleaux and sediment to the bottom, leaving bacteria suspended and able to proliferate in the rocking solution. This process typically completes within 2.5 to 4.5 hours, and the incubation time can be tuned based on target pathogens and their growth rates. Second, to detect positive samples and identify the bacteria, single cell pathogen ID using MB-seqFISH is performed within 1.25 to 2.5 hours (Fig. 1C). This module applies one to three rounds of barcoded nucleic acid probe sets targeting specific regions of bacterial 16S rRNA in individual cells, enabling precise determination of

Gram classification, mono- versus polymicrobial infection, and order-, family-, genus-, and species-level identification from only a small number of bacteria. Third, bacterial cells are encapsulated in gel for single-cell AST (Fig. 1D). A microwell array platform is integrated to trap individual cells and streamline workflow steps that would otherwise be labor intensive, reducing hands-on time and enabling scalable, high-throughput testing. The assay measures temporal changes in cell morphology and growth kinetics to determine susceptibility profiles and minimum inhibitory concentrations (MICs) for multiple antibiotics in approximately 1.5 hours.

STREAM bacterial enrichment from whole blood

Current rapid BSI diagnostic protocols are often limited by isolation efficiency and interference with blood matrix components (15–18). To overcome these limitations, we developed STREAM, a system that integrates rapid bacterial culture and isolation into a single process. The workflow begins by mixing 10 ml of whole blood with a customized culture broth and dextran formulation (~35 ml), followed by incubation at 37°C under gentle orbital shaking at 65 rpm (Fig. 2A and fig. S1A). To accelerate bacterial expansion from low initial loads, we evaluated multiple media formulations across common BSI pathogens (fig. S1, B to D). The brain heart infusion-tryptic soy broth (BHI-TSB) formulation supplemented with Fe^{2+} and Ca^{2+} ions produced the most rapid and consistent outgrowth. The culture broth dilutes and helps neutralize blood-borne inhibitors and provides

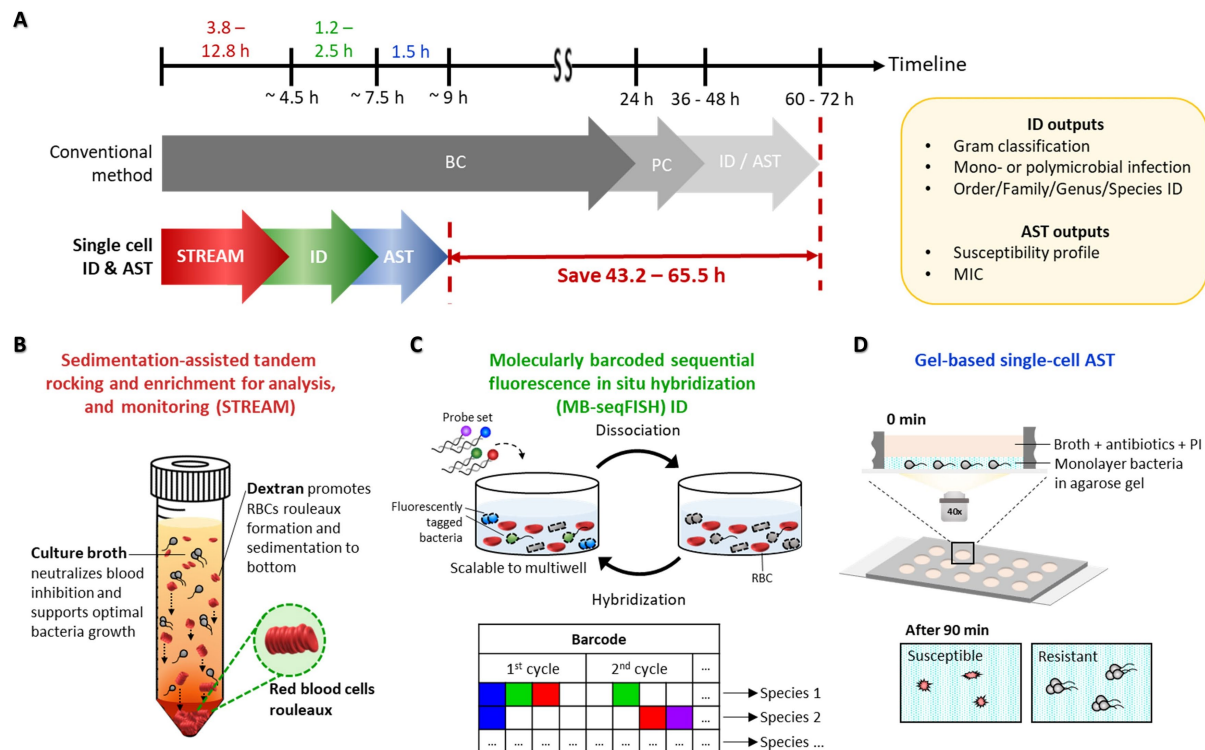


Fig. 1. Rapid and robust BSI diagnosis with STREAM, MB-seqFISH, and gel-based single-cell AST. (A) Diagnostic process for conventional blood culture techniques and our platform, which combines STREAM, single-cell pathogen ID and gel-based AST. BC, blood culture; PC, pure culture. (B) Simultaneous rapid bacterial enrichment and isolation directly from whole blood using a customized dextran-culture broth formulation with gentle rocking in STREAM. RBC aggregates sediment to the bottom while bacteria remain in the plasma-like layer. (C) Pathogen ID workflow for ultralow bacterial load using MB-seqFISH by coupling a molecular barcoding scheme with sequential FISH, and is completed within 1.25 to 2.5 hours (h). (D) Gel-based single-cell AST workflow using gel-immobilized bacteria to assess single-cell growth, morphological responses, and viability for determining susceptibility profiles and MIC values. PI, propidium iodide.

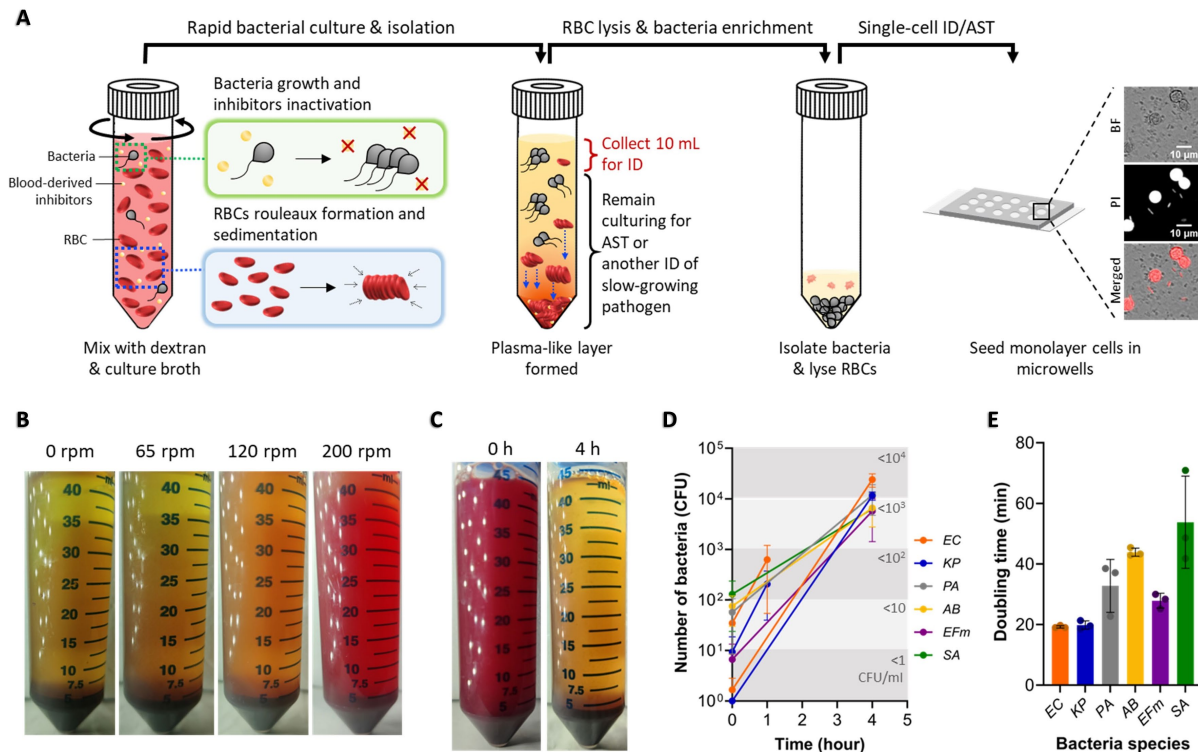


Fig. 2. STREAM for rapid enrichment and isolation of bacteria from whole blood. (A) Workflow of simultaneous rapid bacterial culture and isolation from whole blood. Microscopy images show isolated samples deposited into multiplexed wells for subsequent single-cell ID and AST analysis. (B) Photographs illustrating RBC depletion efficiency in whole blood spiked with *E. coli* (1 CFU/ml) at different orbital shaking speeds after 4 hours of incubation. (C) Photographs of the blood-formulation tube at 0 and 4 hours, showing a bacteria-rich, plasma-like layer at the top and sedimented RBC rouleaux at the bottom. (D) Bacterial recovery from whole blood with initial spiked concentrations of 0.1 to 10 CFU/ml across different time points. (E) Bacterial doubling time during STREAM. Data are presented as mean $\pm$ SEM and $n = 3$. *EC*, *E. coli*; *KP*, *K. pneumoniae*; *PA*, *P. aeruginosa*; *AB*, *A. baumannii*; *EFm*, *E. faecium*; *SA*, *S. aureus*.

essential nutrients to support bacterial growth in agreement with previous reports (27–31). STREAM also incorporates dextran, which induces RBC aggregation through depletion interactions, driving rouleaux formation and gradual sedimentation to the bottom of the container (18, 32). A plasma-like layer enriched with actively proliferating bacteria formed at the top. At regular intervals, the top fraction (10 ml) is sampled to assess culture positivity and perform pathogen ID, while the remaining culture continues to incubate for subsequent AST or additional pathogen detection if needed. Retaining this residual culture enables follow-up testing of slow-growing or antibiotic-exposed pathogens, reducing the risk of false negatives when initial ID is negative or inconclusive.

The efficiency of RBC depletion and bacterial isolation depends on the orbital shaking rate, reflecting the size dependence of settling velocity and the opposing effect of shear-induced re-entrainment. The theoretical basis is provided in Supplementary Text. Experimentally, without shaking, RBC aggregates sedimented effectively; however, bacterial recovery was low because both cell types settled to the bottom. Gentle shaking (e.g., at 65 rpm) produced a clearly separated plasma-like layer comparable to the no-shaking condition (Fig. 2B). Under these conditions, RBCs coalesced into large rouleaux while bacterial cells were re-entrained into the solution. Meanwhile, the orbital motion also enhances oxygen transfer, supporting rapid bacterial growth. Some plasma proteins and immune factors may co-sediment with these aggregates (32). This process yields a nutrient-rich,

plasma-like upper layer with the majority of RBC removed. At higher shaking speeds (e.g., 120 to 200 rpm), the plasma-like layer displayed lower clarity, indicating that some RBCs and smaller rouleaux were entrained into the solution (Fig. 2B). Figure 2C shows a clear separation between the plasma-like layer and the RBC layer after 4 hours of STREAM.

To evaluate the performance of STREAM, human whole blood was spiked with common BSI pathogens, including *Escherichia coli*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, *Enterococcus faecium*, and *Staphylococcus aureus*, at 0.1 to 10 CFU/ml. After 4 hours of incubation, bacterial populations were enriched by two to four orders of magnitude relative to the initial inoculum (Fig. 2D). For fast-growing species such as *E. coli* and *K. pneumoniae*, 1 hour of incubation was sufficient for robust single-cell pathogen ID, whereas 4 hours was adequate for all tested pathogens. Most bacteria expanded at rates comparable to their expected doubling times, except *S. aureus*, which exhibited a noticeably slower apparent growth rate (typical doubling time ~30 to 40 min) (Fig. 2E). *S. aureus* is known to be difficult to recover from blood (17, 18), likely because it agglutinates via clumping factors that bind host proteins, leading to partial entrapment within RBC rouleaux (33). Overall, STREAM enables robust and rapid enrichment of bloodstream pathogens, including challenging species, to concentrations suitable for downstream single-cell analysis.

MB-seqFISH for single cell pathogen ID

To identify diverse BSI pathogens from the STREAM-enriched sample, which may only contain a low number of bacteria, we developed MB-seqFISH, a single-cell pathogen ID technique that couples a molecular barcoding scheme with seqFISH (24). The high abundance of 16S rRNA ($\approx 10^4$ to 10^5 copies per cell) within the small, confined bacterial volume provides a high effective target concentration, generating strong signal-to-noise ratios for seqFISH without amplification. By designing a molecular barcoding scheme, the process rapidly classifies and identifies target bacteria from low-bacterial-load samples. The workflow comprises fixation of bacteria, permeabilization of the bacterial cell wall, iterative probe hybridization and dissociation, and fluorescence imaging in a microwell array (Fig. 3A). Experimental parameters were optimized to maximize detection efficiency across a panel of common bacteria (table S1). Positive samples can be confirmed after the first round of staining. Gram-negative and Gram-positive pathogens can be resolved to species-level within two to three staining rounds.

In each cycle, a distinct set of fluorescent probes targeting bacterial 16S rRNA is introduced into the cells (Fig. 3B). The probes were designed using a custom R package considering the diversity and similarity of the target sequences, the secondary structure of 16S rRNA, affinity of target-probe binding, and specificity of the probe (26). Probes were generally designed to target open-loop regions, which achieved efficient hybridization and showed high signals. Both single-stranded probes and double-stranded probes with a complementary quencher sequence for enhancing the probe specificity were designed (table S2). Single-stranded probes were used for most species to allow rapid hybridization and strong signal intensity, whereas closely related species were resolved using double-stranded probes containing a quencher strand to enhance specificity (Fig. 3, C and D) (34). Multiple probes could be incorporated to ensure complete coverage. For instance, two genus-level probes targeting *Klebsiella* spp. were designed across *Klebsiella* species. The fluorescence barcode guided the selection of the next probe sets until the pathogen was identified. After each round of imaging, the probes were fully dissociated from their targets, resetting the signal for the next round. As shown in Fig. 3E, following each hybridization cycle with the probe, fluorescence signal returned to baseline, confirming complete probe dissociation and enabling successive rounds without residual signal interference.

Figure 3F illustrates the molecular barcoding scheme for pathogen ID. In the first round, a probe set comprising a Universal probe (UNI), a Gram-negative probe (GN), an Enterobacteriaceae + *Serratia marcescens* (Ebceae+SM) probe, and a *Haemophilus influenzae* (HI) probe was applied. On the basis of the resulting color codes, cells were classified into four groups: cells showing blue, green, and far-red signals were identified as *H. influenzae*; cells showing blue, green, and red signals were classified as group 2a, consisting of Gram-negative bacteria from Enterobacteriaceae and *S. marcescens*; cells showing blue and green signals were classified as group 2b, consisting of Gram-negative bacteria excluding *H. influenzae* and group 2a; and cells showing only blue or no signal were classified as group 2c, consisting of Gram-positive, out-of-panel bacteria, or nonbacterial cells. Following this initial classification, a second set of distinctive probes was applied to each group. Species-level ID for Gram-negative bacteria in groups 2a-b was resolved, whereas group 2c was further subdivided: *Staphylococcus* spp. (green and red, group 3a), *Streptococcus* spp. (red and far-red, group 3b), *Enterococcus* spp. or

other Gram-positive bacteria (red, group 3c), and samples requiring retesting for out-of-panel or nonbacteria (no signal). In the third cycle, subsequent application of group-specific probe sets enabled species-level ID of Gram-positive bacteria within groups 3a-3c.

We first evaluated the performance of probes using a panel of 21 bacterial pathogens associated with BSI (Fig. 4A and table S1). The probes generally exhibited high sensitivity, specificity, and detection efficiency exceeding 90%. To validate clinical applicability, MB-seqFISH was validated using positive blood culture samples from BSI patients collected at the Penn State Hershey Medical Center clinical microbiology laboratory. The samples encompassed both Gram-positive and Gram-negative pathogens, including all ESKAPE species (Fig. 4B). MB-seqFISH achieved 96.15% concordance with the clinical microbiology laboratory report, all within 1.2 to 2.5 hours by analyzing at least 50 bacterial cells (Fig. 4C and fig. S2). Four samples (4 of 104, 3.85%) were under-identified, i.e., not fully resolved. Specifically, 1 of 16 *K. pneumoniae* samples and 1 of 21 *E. coli* samples were correctly assigned to the Enterobacteriaceae family but not resolved to the genus or species level. One *Staphylococcus lugdunensis* sample was only classified as Gram-positive, and one *Stenotrophomonas maltophilia* sample was recognized by the UNI probe without further classification. No misidentification errors were observed.

Figure 4D illustrates the detection of a spiked polymicrobial sample containing *P. aeruginosa* and *P. mirabilis* with probe set 2b. *P. aeruginosa* hybridized only with the green PM + PA probes, whereas *P. mirabilis* bound both the green PM + PA and red EBril probes, appearing yellow in the image. Figure 4E shows detection of three positive blood culture samples containing *E. coli* (sample 1), *S. marcescens* (sample 2), and *S. aureus* (sample 3). Samples 1 and 2 were positive for UNI, GN, and Ebceae+SM probe in the first round and were identified by species probes in the second round (probe set 2a). For *S. aureus*, the bacteria hybridized only with the UNI probe in the first round. A genus-level probe in probe set 2c confirmed the genus of the organism, and a third-round SA probe resolved the species. Notably, the barcoding system is designed with built-in redundancy to support identification at higher taxonomic levels (e.g., genus and order) and ensure coverage. For species with higher prevalence or clinical significance, the barcode scheme can be modified to allow direct detection, e.g., by replacing the HI probe in probe set 1 or by adding additional color channels in probe sets 2 and 3. Collectively, these results demonstrate that MB-seqFISH allows rapid, robust, and sensitive identification of BSI pathogens at the single-cell level.

Gel encapsulation for phenotypic single-cell AST

In our workflow, the STREAM-enriched sample is applied for AST after pathogen ID to provide comprehensive BSI diagnostics. We developed a gel-based single-cell AST system that monitors responses of individual bacterial cells to antibiotics exposure, enabling rapid, phenotypic determination of antimicrobial susceptibility and MIC (Fig. 5A). Incorporation of microgel matrix creates discrete microdomains that standardize the inoculum (13, 14), while gel encapsulation enables stable time-lapse imaging of individual bacterial cells during antimicrobial exposure. We integrate multimodal phenotypic readouts, including changes in cell number, morphology, and membrane integrity, to improve both the accuracy and speed of phenotypic AST (25, 35).

The assay is performed in a scalable microwell format that requires minimal sample volume and facilitates multiplexed analysis

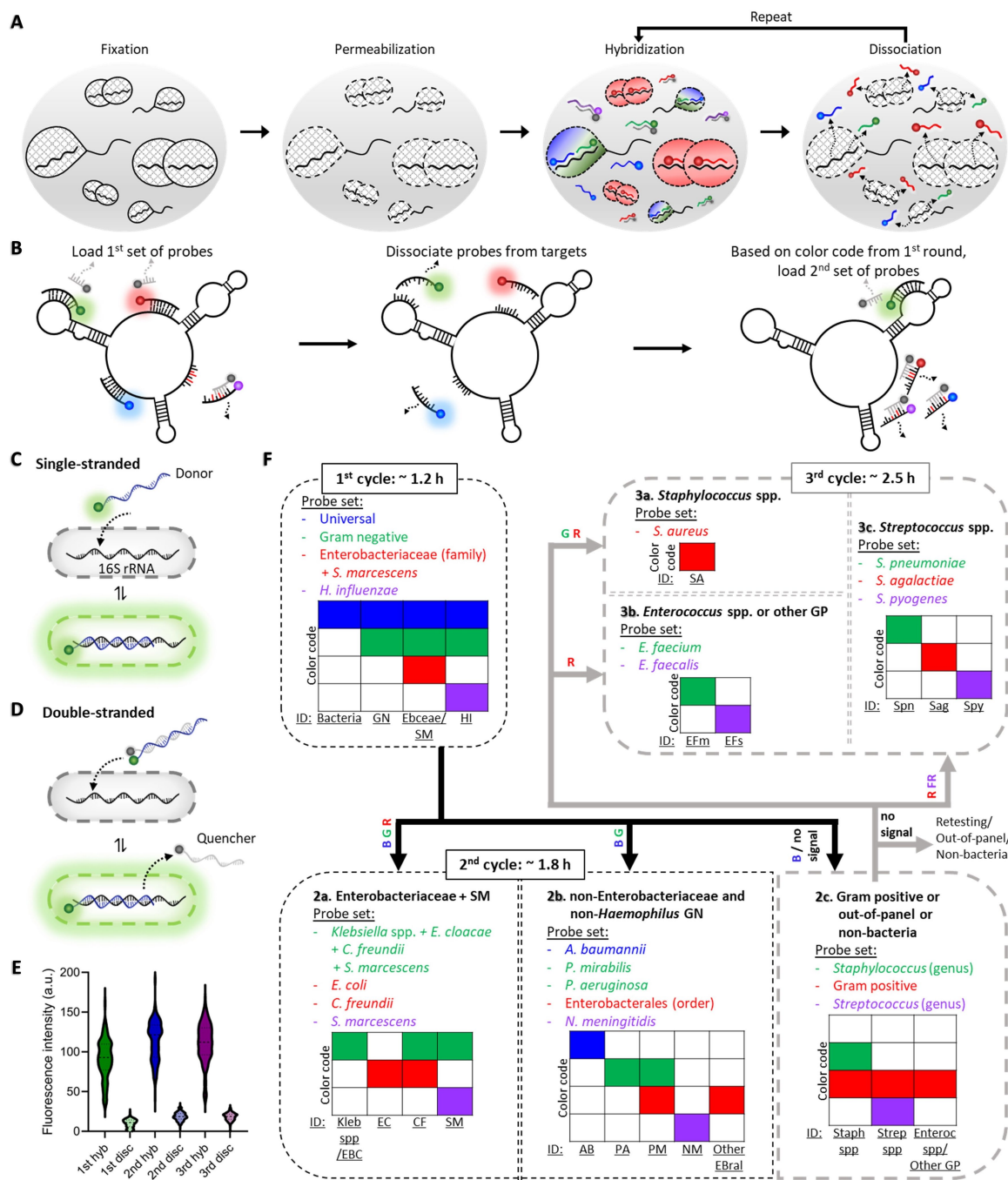


Fig. 3. MB-seqFISH for rapid single-cell pathogen ID. (A) Schematic overview of the MB-seqFISH workflow. Cells are fixed and permeabilized, followed by iterative hybridization and dissociation of fluorescent probes targeting 16S rRNA within individual bacteria for single-cell ID. (B) Multiple probes simultaneously hybridize to distinct accessible loop regions within the 16S rRNA molecule, enabling specific and sensitive detection. After imaging, probes are fully dissociated, and a new probe set is introduced based on the color code from the prior cycle; this process is repeated until the pathogen is identified. (C and D) Design of single- and double-stranded biosensors for 16S rRNA detection. A quencher incorporated in double-stranded probes enhances specificity for distinguishing closely related species. (E) Fluorescence intensity of *K. pneumoniae* across three rounds of hybridization and dissociation using the universal probe, demonstrating efficient probe removal with minimal impact on subsequent hybridization performance. $n = 100$. (F) Molecular barcoding scheme using only four fluorophores to encode a broad pathogen panel. Color codes readouts from each cycle guide probe selection for subsequent hybridization cycles, and unique barcode patterns enable precise pathogen classification and ID.

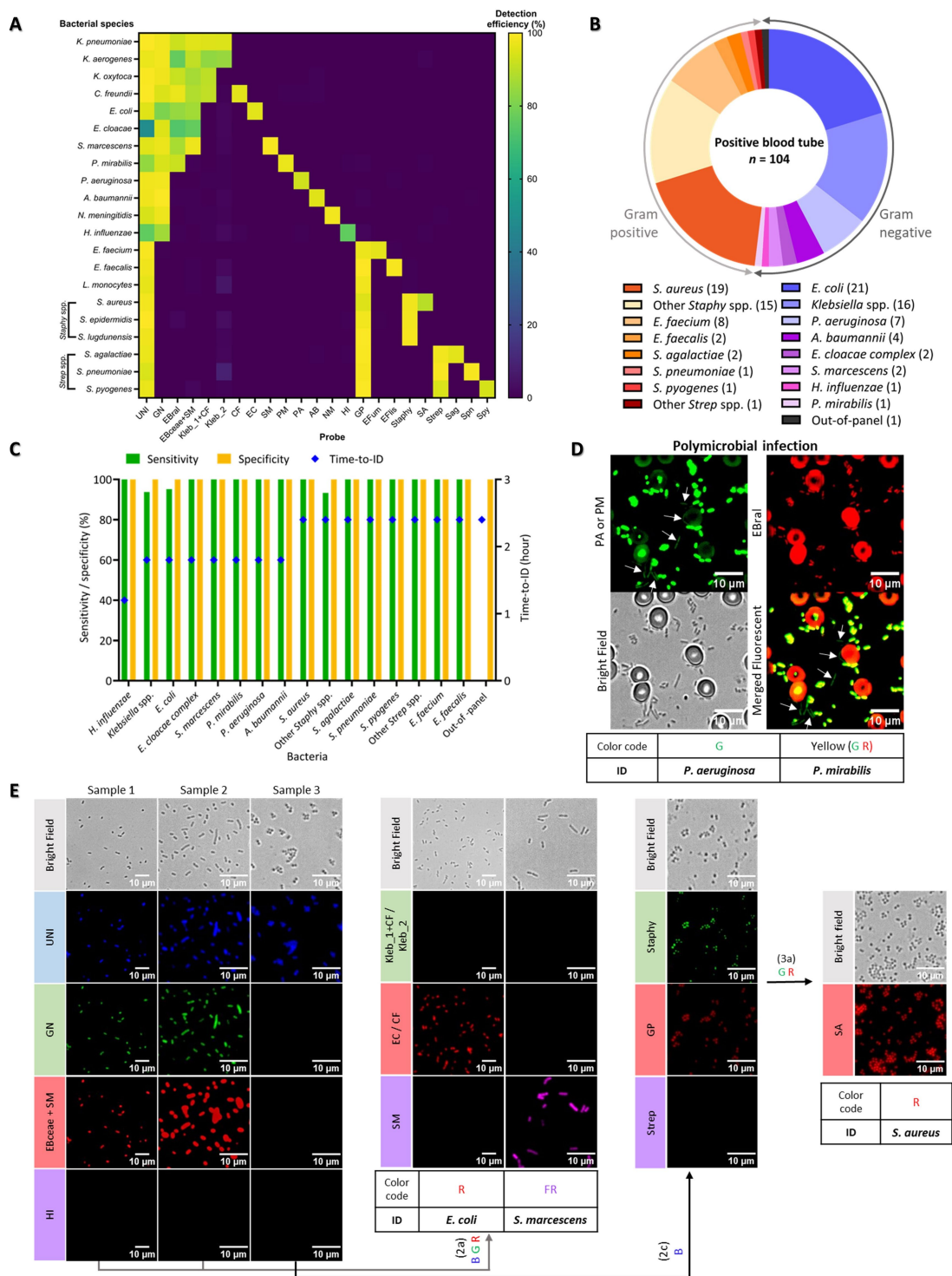


Fig. 4. Characterization and validation of MB-seqFISH. (A) Detection efficiency of probes across 21 most common BSI pathogens. (B) Distribution of bacterial species in positive blood culture samples received from the clinical microbiology laboratory for clinical validation. (C) Sensitivity, specificity, and time-to-ID for clinical positive blood culture samples. (D) Microscopy images of a spiked polymicrobial sample showing *P. aeruginosa* (green signal from PA + PM probes, indicated by arrow) and *P. mirabilis* (yellow, resulting from green PA + PM and red Ebral probes). (E) Microscopy images of three clinical positive blood culture samples identified by MB-seqFISH: Sample 1, *E. coli*; sample 2, *S. marcescens*; and sample 3, *S. aureus*, each resolved by its unique barcode pattern.

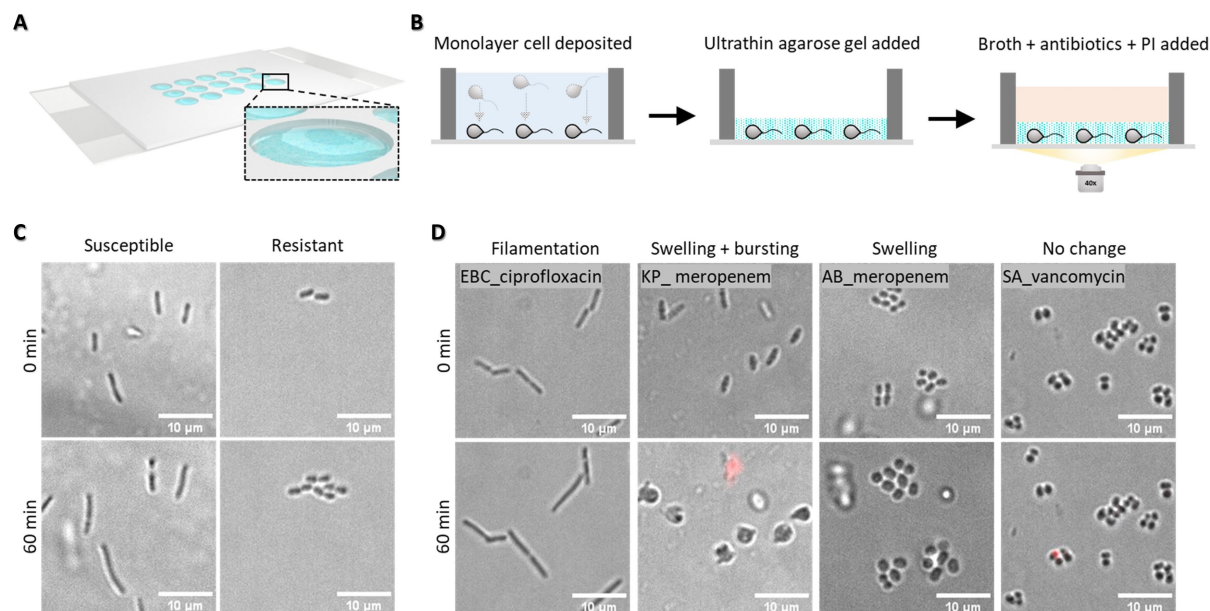


Fig. 5. Gel-based single-cell AST. (A) Illustration of a microfabricated chip featuring a scalable array of microwells for multiplex single-cell AST. (B) Schematic of a single-well unit showing bacterial cells centrifuged onto the glass surface and immobilized in ultralow-melting agarose, followed by incubation with antibiotics of varying types and concentrations for single-cell imaging analysis. (C) Bright-field images of *E. coli* clinical isolates susceptible or resistant to ciprofloxacin (2 $\mu\text{g}/\text{ml}$), displaying distinct morphological features. (D) Examples of morphological responses, including filamentation, swelling with bursting, swelling alone, and no morphological change, observed in bacteria susceptible to antibiotic exposure. Dead cells with disrupted membranes are stained red by PI. Images in the swelling + bursting panel were extracted from Fig. 6G, where the full image sequence is presented.

of multiple drug-dose combinations. The workflow begins with deposition of STREAM-enriched bacterial cells to the bottom of each microwell by centrifugation (Fig. 5B). A thin layer ($\sim 240\ \mu\text{m}$) of ultralow-gelling-temperature agarose at 37°C is then gently overlaid to embed the cells in place. The chip is then cooled for 30 min to allow complete gelation and spatial stabilization of individual bacteria. Culture broth containing the desired concentrations of antibiotics and propidium iodide (PI) is subsequently added to each well, followed by incubation at 37°C . Bacterial growth and viability are monitored continuously by time-lapse microscopy. Figure 5C shows example responses of susceptible and resistant *E. coli* clinical isolates exposed to ciprofloxacin (2 $\mu\text{g}/\text{ml}$). After 60 min, the resistant strain displayed normal bacterial growth dynamics, whereas the susceptible strain exhibited filamentation without cell division. Other morphological alterations were also observed in other susceptible cells, including swelling with bursting, swelling without lysis, and a phenotype showing no cell division and no apparent morphological change (Fig. 5D) (9).

We tracked both the relative number of bacteria and change in area of bacteria clusters for 1 hour. For Gram-negative bacteria, which often undergo filamentation or swelling without division, we primarily analyzed changes in cell number. For Gram-positive species that often form clusters, we quantified changes in cluster area. Loss of membrane integrity, as indicated by PI staining, was used as an additional marker of cell death. Because the high-resolution single-cell morphological analysis can detect minimal increases in cell size (12, 25, 35), even for bacteria that are ultimately susceptible to the antimicrobial, depending on the drug's mechanism and kinetics, we implemented a conservative threshold to avoid false-positive growth calls. In this study, bacterial cells or clusters that exhibited $<60\%$ of the growth observed in untreated controls were classified

as inhibited. Accordingly, the MIC is defined as the lowest antibiotic concentration at which growth is reduced to below 60% of the untreated control.

For example, the ciprofloxacin and meropenem susceptible isolate *Enterobacter cloacae* 041125 showed rapid growth in the absence of antibiotics (Fig. 6, A and B). At 0.25 $\mu\text{g}/\text{ml}$ of ciprofloxacin (MIC), most cells exhibited filamentous morphology and stopped dividing. This filamentation reflects an early phenotypic marker of ciprofloxacin susceptibility, preceding overt cell death. At higher concentrations (0.5 to 2 $\mu\text{g}/\text{ml}$), bacterial growth was completely inhibited. In contrast, exposure to meropenem at concentrations $\geq 1\ \mu\text{g}/\text{ml}$ (MIC) induced extensive cell swelling followed by bursting (Fig. 6, C and D). Increasing meropenem concentrations accelerated cell swelling, ultimately resulting in complete rupture. Figure 6 (E and F) illustrates antimicrobial response of a *K. pneumoniae* isolate (013125) that is resistant to ciprofloxacin and susceptible to meropenem. At 0.25 and 0.5 $\mu\text{g}/\text{ml}$, bacterial proliferation remained comparable to the untreated control. At or above the MIC (1 $\mu\text{g}/\text{ml}$), ciprofloxacin induced filamentous morphology and halted cell replication in most cells, with growth markedly suppressed at 2 $\mu\text{g}/\text{ml}$. In contrast, treatment with meropenem above 1 $\mu\text{g}/\text{ml}$ caused most cells to swell within 30 minutes and partially burst by 60 min (Fig. 6, G and H). Meropenem's effect was further supported by PI staining, which indicated loss of membrane integrity in affected cells. These observations are consistent with MIC values reported by clinical microbiology laboratories. Additional examples are provided in fig. S3.

To assess the clinical performance of the gel-based single-cell AST, we tested a panel of BSI pathogens isolated from positive blood vials, including all ESKAPE bacteria, using antibiotics and concentrations according to CLSI guidelines (table S3). The results are

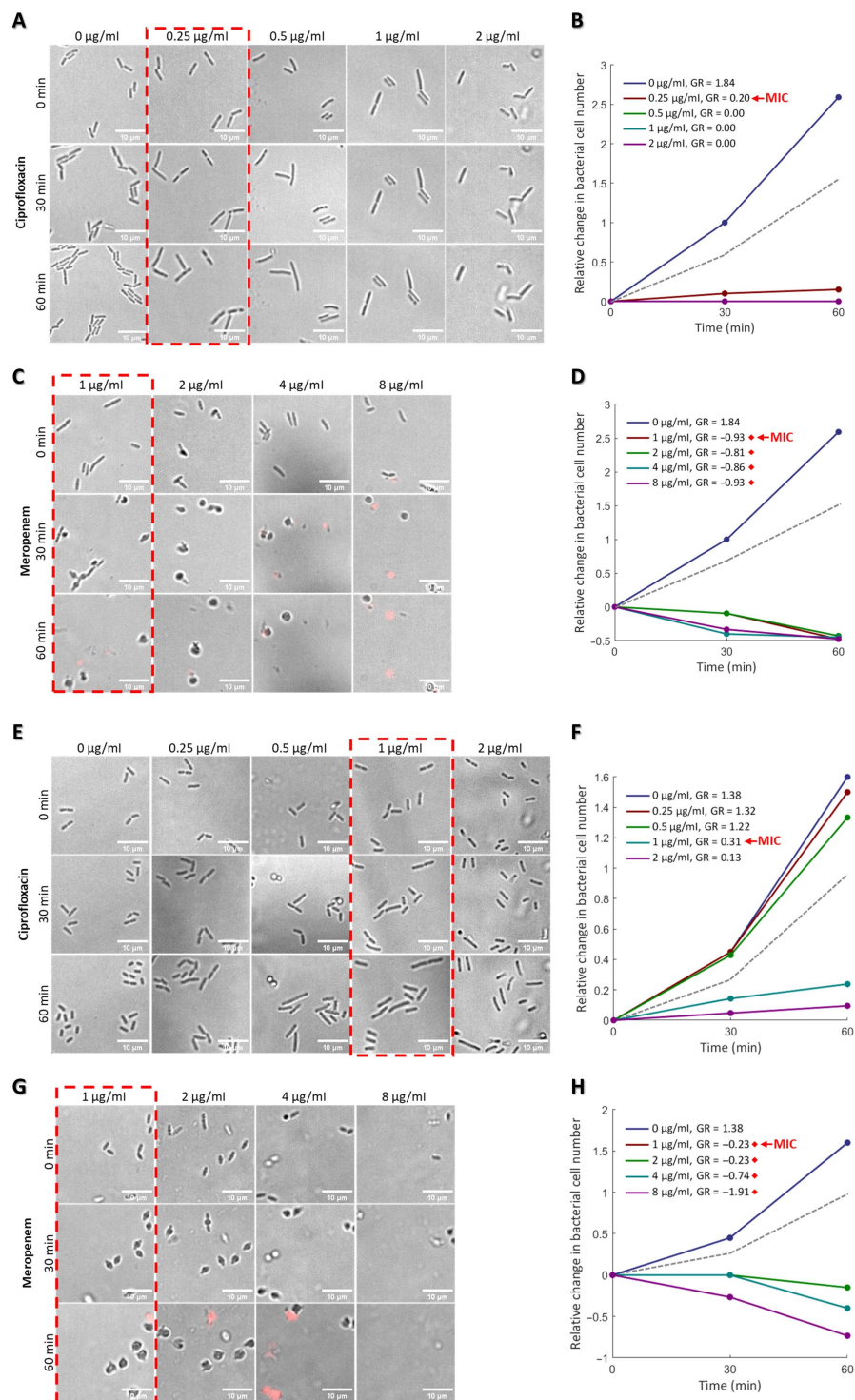


Fig. 6. Gel-based single-cell AST of *E. cloacae* 041125 and *K. pneumoniae* 013125 clinical isolates. (A to D) Time-lapse merged bright-field and propidium iodide (PI) images and relative changes in bacterial cell number showing responses of *E. cloacae* 041125 to ciprofloxacin (A and B) and meropenem (C and D). (E to H) Corresponding time-lapse images and cell-number changes showing responses of *K. pneumoniae* 013125 to ciprofloxacin (E and F) and meropenem (G and H). The dotted gray lines indicate the resistance threshold. Red diamonds indicate the presence of dead cells by PI staining; MIC values are shown with red dotted boxes or arrows. $n = 20$ for all bacteria-drug combinations. Portions of the image in (G) were reused in Fig. 5.

compared with the results obtained from the clinical microbiology laboratory. In total, 219 drug-dose combinations were tested using the gel-based single cell AST. During a 1-hour antibiotic exposure, ~20 individual Gram-negative cells were tracked for relative changes in number, while the area of 10 Gram-positive bacterial clusters was monitored. The MICs for were generally within one twofold dilution of the clinical microbiology laboratory reports (Fig. 7 and table S4). Overall, we achieved an essential agreement of 97.96% and a categorical agreement of 93.9%, with 4.1% very major error, 2.04% minor errors, and no major errors, supporting the use of the gel-based single cell AST for BSI diagnostics.

Complete BSI diagnosis with ultra-low loads blood

To evaluate the speed and robustness of our platform for complete BSI diagnosis, human whole blood was spiked with BSI clinical isolates (Fig. 8A). For *K. pneumoniae* spiked at 1 CFU/ml, bacteria became detectable after 2 hours of STREAM, and pathogen ID was completed at 4.82 hours. Once Gram-level information was obtained, the remaining culture was collected for AST. AST was finished in under 2 hours, yielding comprehensive results at a total turnaround time of 6.75 hours. In contrast, *S. aureus* at 1 CFU/ml was not detected in the first collection after 4 hours of STREAM, likely due to reduced recovery efficiency caused by bacterial agglutination with RBCs. The sample was tested again after 8 hours of STREAM, at which point positivity and species-level ID were achieved at 11.5 hours, and AST was completed by 14 hours. At a higher concentration of *S. aureus* (10 CFU/ml), a 4-hour incubation was sufficient, and ID and AST were completed at 7.5 and 10 hours, respectively.

While patients with BSI can present with bacteria loads between 1 and 10 CFU/ml, quantitative blood culture reveals that the bacteria counts are frequently below 1 CFU/ml (20–23). We, therefore, evaluated the platform using samples with ultralow bacterial loads (~0.1 CFU/ml). With extended STREAM processing (4 hours), the turnaround time for comprehensive ID and AST of *K. pneumoniae* at 0.1 CFU/ml was 8.75 hours. For *S. aureus* at 0.1 CFU/ml, the most challenging case in this study, bacteria were not detectable after 4 hours or even 8 hours of STREAM. Nevertheless, culture positivity could be detected after 11 hours of STREAM, and full pathogen ID and AST were successfully completed with a sample-to-result time of 17 hours. The ability to detect and analyze samples with ultralow bacterial loads is enabled by the single-cell analysis modules, which require only a small number of cells. As shown in Fig. 8 (B and C), MB-seqFISH provided robust, high-signal-to-noise pathogen detection and species-level ID despite background cells and debris.

Likewise, gel-based single-cell AST was performed using only a few bacterial cells or clusters (Fig. 8, D and E). The presence of RBCs did not produce observable effects on bacterial growth, morphology, or viability, and MICs and susceptibility categories were correctly determined (figs. S4 to S5).

DISCUSSION

In this study, we demonstrate that combining STREAM with single-cell analysis enables rapid and robust BSI diagnostics, even for difficult-to-isolate species and samples with ultra-low bacterial concentrations. Although several emerging technologies can isolate bacteria directly from whole blood (15–19), their detection limits typically plateau around ~5 CFU/ml due to the complex blood matrix and limited isolation efficiency, particularly when challenging pathogens are included. As a result, despite its slow turnaround, conventional blood culture remains essential in clinical workflows because of its robustness and sensitivity. STREAM bridges this gap by combining the speed of direct isolation approaches with the diagnostic reliability of blood culture. It depletes RBCs while enriching bacteria in a nutrient-rich, plasma-like layer, effectively coupling rapid culture with size-based isolation. This promotes near-native bacterial growth and yields sufficient cells for downstream analyses even at ultralow loads (0.1 to 1 CFU/ml). STREAM preserves residual culture medium for interval sampling, analogous to blood culture systems with CO₂ monitoring, reducing the likelihood of false negatives due to imperfect isolation or slow-growing bacteria. This time-saving advantage may be particularly pronounced for slower-growing bacteria, for which the higher sensitivity of single-cell detection can reduce the culture time needed to reach a detectable bacterial abundance. Using this combined workflow, the average time-to-result for complete BSI diagnosis directly from whole blood was approximately 9 hours (range: 6.75 to 17 hours), far faster than the multiday timeline required by standard blood culture.

By incorporating single-cell analysis technologies, the platform rapidly performs pathogen ID using STREAM-enriched samples, even when only a small number of bacteria are present. Leveraging recent advances in sequential labeling and molecular barcoding for high-content image-based analysis (24), MB-seqFISH enables a scalable approach for single-cell pathogen identification. A major challenge in rapid molecular diagnostics, such as polymerase chain reaction, is that low bacterial load and the complex blood matrix reduce the signal-to-noise ratio and compromise assay robustness (10). In contrast, seqFISH exploits the high abundance of 16S rRNA

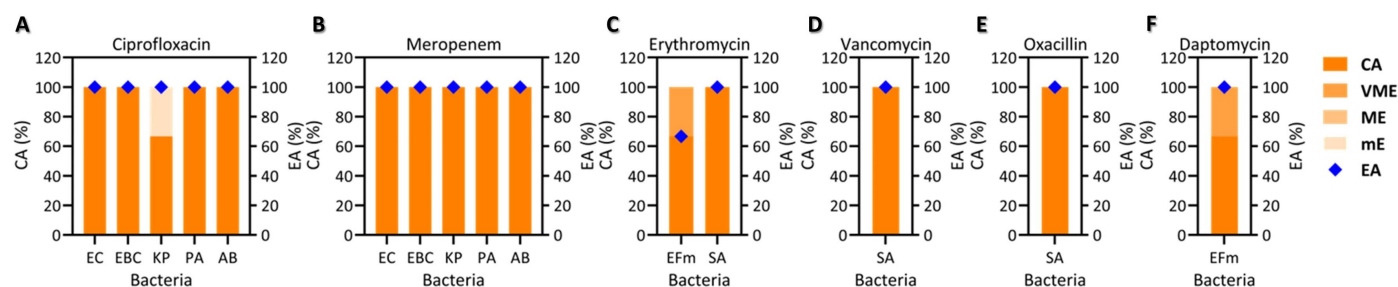


Fig. 7. Performance evaluation of gel-based single-cell AST platform using positive blood culture isolates from BSI patients. (A to F) Categorical agreement (CA) and essential agreement (EA) of AST results compared with the reference method for each antibiotic: ciprofloxacin (A), meropenem (B), erythromycin (C), vancomycin (D), oxacillin (E), and daptomycin (F).

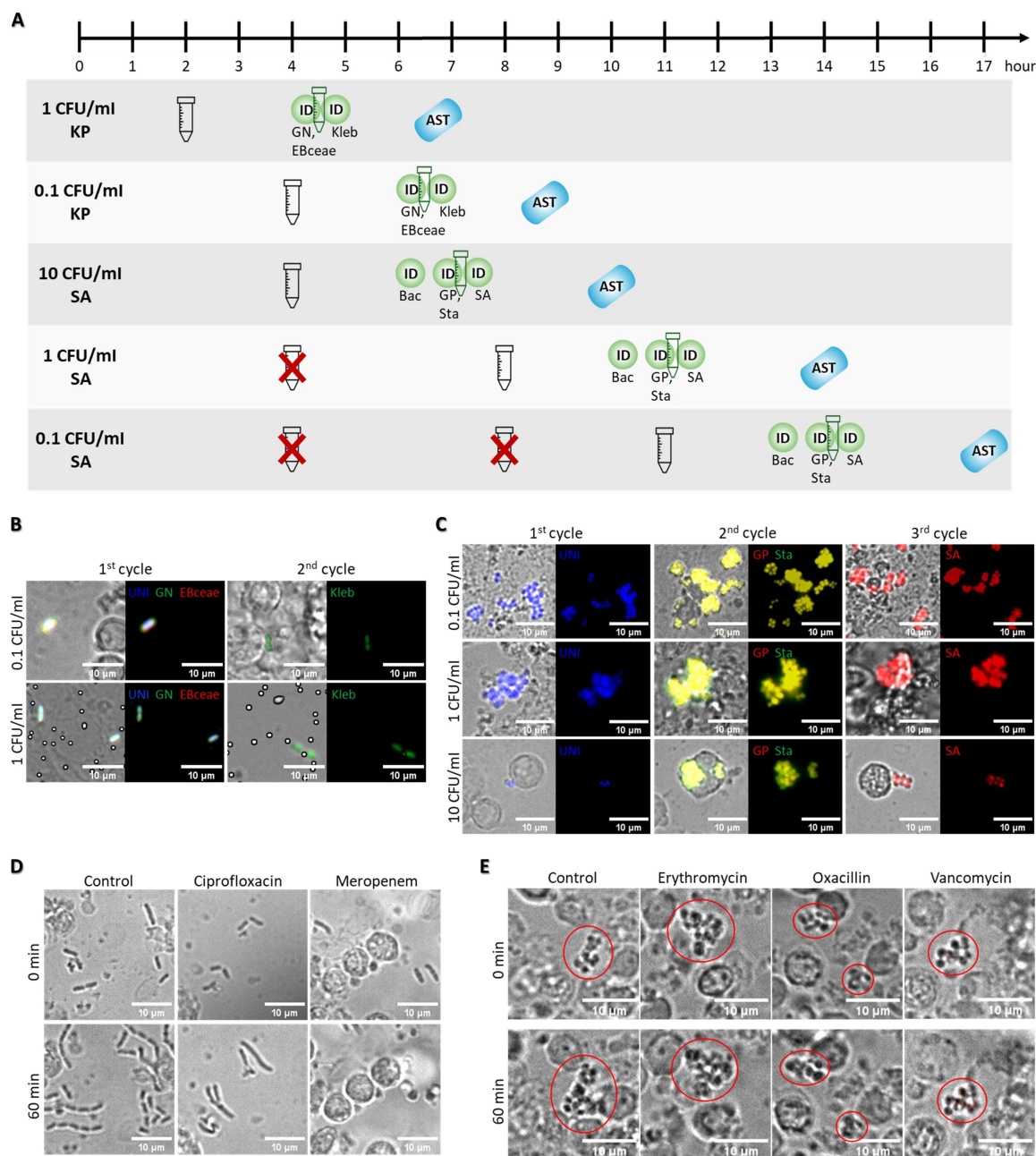


Fig. 8. Performance evaluation of the platform using whole blood spiked with clinical isolates. (A) Timeline of complete BSI diagnosis from whole blood spiked with 0.1 to 10 CFU/ml of clinical isolates *K. pneumoniae* 013125 and *S. aureus* 032725. Black tube: top 10-ml plasma-like layer collected; green tube: remaining plasma-like layer collected; red cross: no bacteria detected in the collected sample. GN, Gram-negative; EBceae, Enterobacteriaceae; Kleb, *Klebsiella* spp.; Bac, bacteria; GP, Gram-positive; Sta, *Staphylococcus* spp.; SA, *S. aureus*. (B and C) Microscopy images of pathogen ID after each probe hybridization cycle of *K. pneumoniae* and *S. aureus*, respectively. (D and E) Bacterial responses at 0 and 60 min under 1x MIC antibiotics. Images in (D) were extracted from fig. S4. Images in (E) were extracted from fig. S5.

in intact bacterial cells and achieves a markedly higher signal-to-noise ratio for single-cell analysis (36). Molecularly barcoded probes facilitate bacterial classification with a minimal number of staining rounds and allow identification across multiple taxonomic levels, providing broad coverage even when species-specific probes are unavailable and help to inform treatment decisions. MB-seqFISH is modular and scalable and can be easily adapted to additional bacterial panels, balancing turnaround time, cost-effectiveness, and the

diversity of BSI pathogens. Given the flexibility of the platform, the assay can also be expanded to detect fungal and anaerobic pathogens with minor modifications, such as altering the culture formulation and conducting experiments in an anaerobic chamber. It provides real-time, actionable clinical information and enables early AST with a relevant drug panel before a specific taxonomic ID is obtained.

For AST, the gel-based single-cell assay rapidly evaluates the phenotypic effects of antimicrobials on BSI pathogens. Our platform

integrates growth kinetics, morphological analysis, and membrane integrity assessment to enhance both accuracy and speed. Prior works using microfluidics and microscopy have demonstrated high-accuracy single-cell AST, enabling rapid MIC determination through microfluidic trapping and real-time imaging (12, 25). By incorporating machine learning, these methods can deliver AST results on the timescale of a single doubling cycle (35). Building on these developments, our single-cell imaging platform combines high-resolution imaging with membrane-integrity measurements using PI staining, providing direct information on cell lysis and enabling detailed time-kill analysis. Notably, conventional AST focuses solely on growth inhibition and does not capture bacterial killing kinetics or clearance dynamics. Given the severe consequences of sepsis, selecting antimicrobials with rapid bactericidal activity may be especially valuable for optimizing BSI treatment, particularly in high-risk or immunocompromised patients.

Several limitations of the single-cell analysis workflow remain to be addressed before clinical translation. Many patients with suspected sepsis receive empiric antibiotics before blood collection. Prior antimicrobial exposure may reduce viable bacterial counts or alter growth kinetics, increasing the risk of false-negative culture results and affecting enrichment efficiency. Incorporating antibiotic-adsorbing blood culture bottles with antibiotic-neutralizing materials into the STREAM process may help mitigate this limitation. Furthermore, polymicrobial infections present diagnostic challenges because STREAM and other culture systems tend to favor fast-growing species that can mask coinfecting organisms. In clinical contexts where polymicrobial infection is common, such as open wounds, intra-abdominal sources, or disrupted epithelial barriers, STREAM incubation combined with MB-seqFISH should be continued even after an initial pathogen is detected to reduce the risk of incomplete or biased detection. In addition, intracellular bacteria, viable but nonculturable cells, and pathogens requiring specialized culture conditions may present challenges for culture-based workflows. Further optimization of culture conditions, including the potential use of selective lytic agents, may be required to improve recovery of these organisms.

Another important consideration is the cost and complexity of the assay. The single-cell analysis workflow involves multiple processing steps, including interval sampling and MB-seqFISH, which may increase cost and operational complexity in clinical microbiology laboratories and potentially hinder translation (10). Future efforts should therefore prioritize workflow simplification and automation. Blood should be collected directly into specialized culture tubes preloaded with appropriate culture medium and a compatible anticoagulant. Integration with automated liquid handling, scalable image analysis pipelines, and AI-driven computer vision will reduce manual intervention and improve cost-effectiveness. In the current design, pathogen identification and AST are performed using custom multi-well arrays compatible with microfluidic systems and robotic handlers, facilitating future automation. Integration with laboratory information systems and clinical decision-support tools will further streamline reporting and support precision antimicrobial therapy. Multicenter clinical studies will be essential to validate performance across diverse patient populations and pathogen spectra. With continued optimization and rigorous clinical validation, the combined STREAM and single-cell analysis workflow may enable next-generation rapid and robust diagnostics for precision management of BSIs (1, 4, 5).

MATERIALS AND METHODS

Experimental design

This study was designed to evaluate the analytical performance and clinical feasibility of our integrated STREAM and single-cell analysis platform for rapid, robust, and comprehensive BSI diagnostics, including pathogen ID and phenotypic AST. We first conducted analytical validation using well-characterized reference bacterial strains. We then evaluated the platform using positive blood culture specimens and clinical isolates obtained from patients with BSI to assess diagnostic concordance with standard-of-care microbiological methods.

Bacterial strains and clinical samples

Standard bacterial strains, listed in table S1, were purchased from the American Type Culture Collection or obtained from Stanford University. Deidentified positive blood culture samples were obtained from Penn State Milton S. Hershey Medical Center. The procedure was approved by the Pennsylvania State University Institutional Review Board (CR00024734). Positive blood cultures samples and isolates were stored in 25% glycerol at -80°C . Human blood samples anticoagulated with citrate phosphate dextrose were purchased from BioIVT and stored at 4°C until use.

Bacterial isolation and enrichment using STREAM

Dextran (500 kDa; Spectrum D1004) and sodium polyanethole sulfonate (SPS) solutions (Sigma-Aldrich) were filtered through a $0.2\text{-}\mu\text{m}$ polyethersulfone membrane before use. Six standard bacterial isolates (*E. coli*, *K. pneumoniae*, *P. aeruginosa*, *A. baumannii*, *E. faecium*, and *S. aureus*) were cultured in Mueller-Hinton II (MH2) broth (Sigma-Aldrich) overnight at 37°C with shaking at 130 rpm under aerobic conditions. Bacteria were subcultured in fresh MH2 broth 1 hour before each experiment and adjusted to a concentration of 10^7 CFU/ml, as quantified using a NanoDrop2000 spectrophotometer (Thermo Fisher Scientific). Bacterial suspensions were then serially diluted, and $40\text{ }\mu\text{l}$ of desired dilution was spiked into 10 ml of human whole blood to achieve final concentrations ranging from 0.1 to 10 CFU/ml. Concurrently, equal volumes of bacterial suspension were spread directly onto CHROMagar ($n = 3$) to backtrack the number of bacterial cells inoculated in the blood. Then, 34.27-ml customized formulation was added to the spiked blood sample, consisting of 1% dextran, 10 ml of brain heart infusion broth (Becton Dickinson), 10 ml of tryptic soy broth (Becton Dickinson), 0.05% SPS, 1.7 mM calcium chloride (Thermo Fisher Scientific), 120 μM iron (II) sulfate (Sigma-Aldrich), and 2 μM argatroban (Sigma-Aldrich). The mixture was incubated at 65 rpm at 37°C . The plasma-like layer (~ 40 ml) formed at the top was carefully collected and transferred to a new 50 ml tube, followed by centrifugation (Thermo Fisher Scientific, Sorvall ST8R) at 3260 rpm for 15 min at 37°C . The resulting pellet was resuspended in 9 ml of $1\times$ RBC lysis buffer (BioLegend), incubated for 15 min at 37°C , and centrifuged again (3260 rpm, 10 min, 37°C). The resulting pellet was resuspended in 100 μl of MH2 broth and plated on CHROMagar to quantify the total number of cells isolated. Doubling time were calculated using the following formula

Time taken from incubation to plating

$$\log_2 \left(\frac{\text{Number of isolated bacterial cells}}{\text{Number of spiked bacterial cells}} \right)$$

To minimize potential bias from donor variability, blood from at least two independent donors was used to characterize each bacterial strain.

Microwell array fabrication

We used a VLS 3.5 Desktop Series laser machining system (Universal Laser Systems) to fabricate microwells on 3-mm-thick acrylic plates and double-sided adhesive tape (3M 468MP). The patterned acrylic and tape layers were then assembled onto glass coverslips (Mercedes Scientific) to create substrates for single-cell microscopy. For pathogen ID, the microwell array consisted of a 4 × 8 layout with a diameter of 7 mm per well, and coverslips were pre-coated with poly-L-lysine (Sigma-Aldrich). For AST, a 3 × 5 microwell array with a diameter of 4 mm per well was fabricated with noncoated coverslips.

Probe design

All probes used in this study are listed in table S2. Published probes reported in previous studies were first evaluated *in vitro* using FISH for performance. (37–42) Probes that exhibited suboptimal performance, or had not been previously developed, were newly designed using a custom R package described in earlier work (26). In brief, 16S rRNA sequences were obtained from the Ribosomal Database Project and the National Center for Biotechnology Information (NCBI), aligned using Clustal Omega (EMBL-EBI), and then imported into R package for probe design. For probes showing low specificity, a quencher strand was added, and the sensitivity-specificity balance was optimized by adjusting the quencher length, where longer quenchers increase specificity but may reduce sensitivity. Donor probes were designed to be 18 to 23 nucleotides with 45 to 60% GC content, while quenchers were 11 to 16 nucleotides in length. Candidate probes were ranked based on the number of nucleotides binding to accessible loops in the 16S rRNA secondary structure, as well as thermodynamic parameters including ΔG_{target} , $\Delta G_{\text{nontarget}}$, $\Delta G_{\text{quencher}}$, $\Delta G_{\text{hairpin}}$, and $\Delta G_{\text{self-dimer}}$. Probes were synthesized by Integrated DNA Technologies (IDT), with donor fluorophores (Texas Red, Alexa Fluor 350, 488, 546, or 647) and quencher fluorophores (Iowa Black FQ or RQ). The optimal probes were selected based on *in silico* validation using the NCBI BLAST tool and *in vitro* performance assessed by FISH using standard bacterial strains. For each probe, signal thresholds were optimized to balance specificity and detection efficiency. Thresholds were established by analyzing the maximum autofluorescence or nontarget signal intensity and the background signal. Detection efficiency for each probe was calculated as the percentage of fluorescently labeled cells relative to the total number of cells measured.

Pathogen ID with MB-seqFISH

Hybridization buffer was prepared with 0.9 M sodium chloride (Sigma-Aldrich), 0.02 M tris hydrochloride (tris-HCl) at pH 7.6 (Sigma-Aldrich), 20% formamide (Sigma-Aldrich), 10% dextran sulfate (Sigma-Aldrich), 0.01% SDS (Sigma-Aldrich), and 0.45 mM calcium chloride. Washing buffer was made up of 0.4× saline-sodium citrate (Sigma-Aldrich). Dissociation buffer was prepared with 0.02 M tris-HCl, 90% formamide, 0.01% SDS, and 5 μM ethylenediaminetetraacetic acid (Sigma-Aldrich). Enzyme buffer was prepared with lysozyme (15 mg/ml; Sigma-Aldrich) and lysostaphin (0.5 mg/ml; Sigma-Aldrich) dissolved in 10 mM tris-HCl at pH 8.0. A 3- μl aliquot of each positive blood culture sample was added into individual wells and incubated with 20 μl of 4% paraformaldehyde for 40 min at

4°C for fixation. Samples were centrifuged at 2560 rpm for 10 min at 37°C, and the liquid was discarded. The chip was briefly passed through a flame to evaporate residual liquid and then washed twice with 80 μl of 1× phosphate-buffered saline (PBS). To permeabilize the cells, 150 μl of 1× PBS was added and incubated for 10 min at 80°C, followed by rapid cooling on ice for 1 min to halt further lysis. Cells were then incubated with 80 μl of hybridization buffer containing 1 μM of specific probe sets at 46°C for 20 min. After hybridization, samples were washed with 80 μl of washing buffer and incubated for 2 min at 46°C, followed by an additional washing step to remove excess probes and reduce nonspecific binding. Fluorescence and bright-field images were acquired using a Leica HCX PLAN APO 40×/0.85 objective lens and fluorescence filter cubes of 4',6-diamidino-2-phenylindole, fluorescein isothiocyanate, Y3, and Y5. Probe dissociation was performed using dissociation buffer at 46°C for 10 min. For Gram-negative bacteria, the second round of hybridization was performed directly after dissociation. For Gram-positive bacteria, samples were first incubated with 80 μl of enzyme buffer for 5 min at 37°C to digest the thicker peptidoglycan cell wall, washed twice with 1× PBS, and then subjected to the next hybridization round. Multiple iterative cycles of probe hybridization, imaging, and dissociation were conducted until pathogen ID was achieved.

Image analysis was performed using a custom ImageJ macro. Bright-field raw images were converted to 8 bit, and background noise was removed using a rolling-ball subtraction algorithm with a 15-pixel radius, assuming a light background. Thresholding was applied using the default ImageJ algorithm, and particles were identified using the “Analyze Particles” function with a minimum particle size cutoff of 0.02 pixels to exclude background artifacts. A watershed segmentation step was applied to separate individual cells, which were then converted into binary mask images with regions of interest (ROIs) recording the location of each cell. Using these ROIs, fluorescence intensity was extracted across all channels, and thresholds were applied to determine probe positivity or negativity. Sensitivity and specificity of the platform for each ID were determined as

$$\text{Sensitivity} = \frac{\text{TP}}{(\text{TP} + \text{FN})}$$

$$\text{Specificity} = \frac{\text{TN}}{(\text{TN} + \text{FP})}$$

Gel-based single-cell AST

A 4% ultralow-gelling-temperature agarose (Sigma-Aldrich) solution was prepared in 1× PBS and heated in a microwave until fully dissolved, then cooled to 37°C in a water bath to avoid thermal damage to the cells. Antimicrobial solutions were purchased from Sigma-Aldrich and prepared according to manufacturer instructions and CLSI guidelines, and 1.5 μl of 10 mM PI dye (Thermo Fisher Scientific) was added per milliliter of each antibiotic solution. *E. coli*, *K. pneumoniae*, *P. aeruginosa*, and *A. baumannii* were tested with ciprofloxacin and meropenem; *E. faecium* with erythromycin and daptomycin; and *S. aureus* with oxacillin, vancomycin, and erythromycin. Clinical isolates from Penn State Milton S. Hershey Medical Center were cultured in MH2 broth overnight at 37°C with shaking at 130 rpm under aerobic conditions. Bacteria were subcultured in fresh MH2 broth 1 hour before each experiment to ensure cells were harvested in the exponential growth phase. Bacteria were adjusted to 10⁵ CFU/ml, and 3 μl of suspension (~10² CFU to mimic

the post-STREAM concentration) was added to individual microwells, followed by centrifugation at 2560 rpm for 10 min. After removal of the supernatant, 3 μ l of agarose gel was added to each well, and waited for 30 min to allow complete gelation and stabilization of cell positions. Antibiotic solutions of varying types and concentrations were added to each microwell and incubated at 37°C. Bacterial growth was monitored via bright-field and fluorescence microscopy using a Leica HCX PLAN APO 40 $\times$ /0.85 objective lens. Bright-field images were obtained with a 15 ms exposure, while PI fluorescence images were collected using a Leica Y3 filter cube with a 50 ms exposure time.

Image analysis was performed using a custom ImageJ macro. Raw images were converted to 8 bit, and background noise was removed using a rolling-ball subtraction algorithm with a 15-pixel radius, assuming a light background. Thresholding was applied using the default ImageJ algorithm, and particles were identified using the Analyze Particles function with a minimum particle size cutoff of 0.02 pixels to exclude background artifacts. For Gram-negative bacteria, a watershed segmentation step was applied before data extraction to separate individual cells. Morphological features, including area, mean fluorescence intensity, and perimeter, were extracted for each identified particle. MATLAB scripts were used to compute and plot the average relative changes in cell number (Gram-negative) or cluster area (Gram-positive) over time. For Gram-negative bacteria, the growth rate was calculated as the difference between the logarithm of the cell count at 60 min and at 0 min, divided by the logarithm of 2 per hour. For Gram-positive bacteria, the growth rate was determined by averaging the change in cluster area between 0 and 60 min over time. The SEM for Gram-positive cluster area change was calculated as the SD divided by the square root of the number of clusters analyzed. PI-positive cells were identified by applying an intensity threshold (20 a.u.) to the mean fluorescence signal. Antimicrobial susceptibility was determined by comparing the growth rate to untreated controls, with $\geq 60\%$ of control growth classified as resistant and $< 60\%$ classified as susceptible. The overall performance was evaluated using these formulas

Essential agreement =

$$\left(\frac{\text{Number of isolates with MIC within } \pm 1 \text{ dilution}}{\text{Total number of isolates tested}} \right) * 100$$

Categorical agreement =

$$\left(\frac{\text{Number of isolates with same category (S/I/R)}}{\text{Total number of isolates tested}} \right) * 100$$

Very Major Error (False susceptible) =

$$\left(\frac{\text{Number of VME}}{\text{Number of resistant isolates by reference}} \right) * 100$$

Major Error (False resistant) =

$$\left(\frac{\text{Number of ME}}{\text{Number of susceptible isolates by reference}} \right) * 100$$

Minor Error (Reference I, test is S/R) =

$$\left(\frac{\text{Number of mE}}{\text{Total number of isolates tested}} \right) * 100$$

Software

All figures were created using Microsoft PowerPoint (version 16). Data and statistical analyses and plotting were carried out using Microsoft Excel (version 16), MATLAB (version R2022b) and GraphPad (version 2024). All image-processing algorithms were developed using ImageJ and MATLAB (version R2022b).

Statistical analysis

All assays were performed in triplicate unless specified otherwise. Data are presented as mean $\pm$ SEM.

Supplementary Materials

This PDF file includes:

Supplementary Text

Figs. S1 to S5

Tables S1 to S4

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